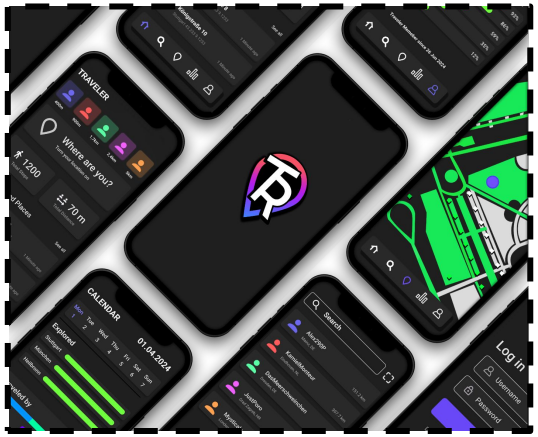
RAMAJANA
SKOPLJAK

SOFTWARE
PORTFOLIO

2025
26TH FEBRUARY

TRAVELER

TECHNOLOGIES & PLATFORM: FLUTTER & UNITY, FOR ANDROID
DESCRIPTION: BACHELOR'S THESIS PROJECT THAT USES OPEN STREET MAPS TO CREATE A 3D WORLD WITH REAL TIME TRACKING
ROLE AND CONTRIBUTION: SOLO PROJECT, RESPONSIBLE FOR ALL DEVELOPMENT AND DESIGN
OUTCOME: PROTOTYPE FOR ANDROID USERS
TIME PERIOD: FEBRUARY 2024 - JUNE 2024





FOREST

TECHNOLOGIES & PLATFORM: FLUTTER, FOR ANDROID
DESCRIPTION: ANDROID APP TO MONITOR VINEYARDS
ROLE AND CONTRIBUTION: TEAM PROJECT, TEAM LEADER, RESPONSIBLE FOR MOST DEVELOPMENT AND DESIGN
OUTCOME: PROTOTYPE FOR ANDROID USERS - SOLO PRESENTED AS HEAD OF MY STUDY PROGRAM
TIME PERIOD: APRIL 2023 - JUNE 2023

TRAVELER

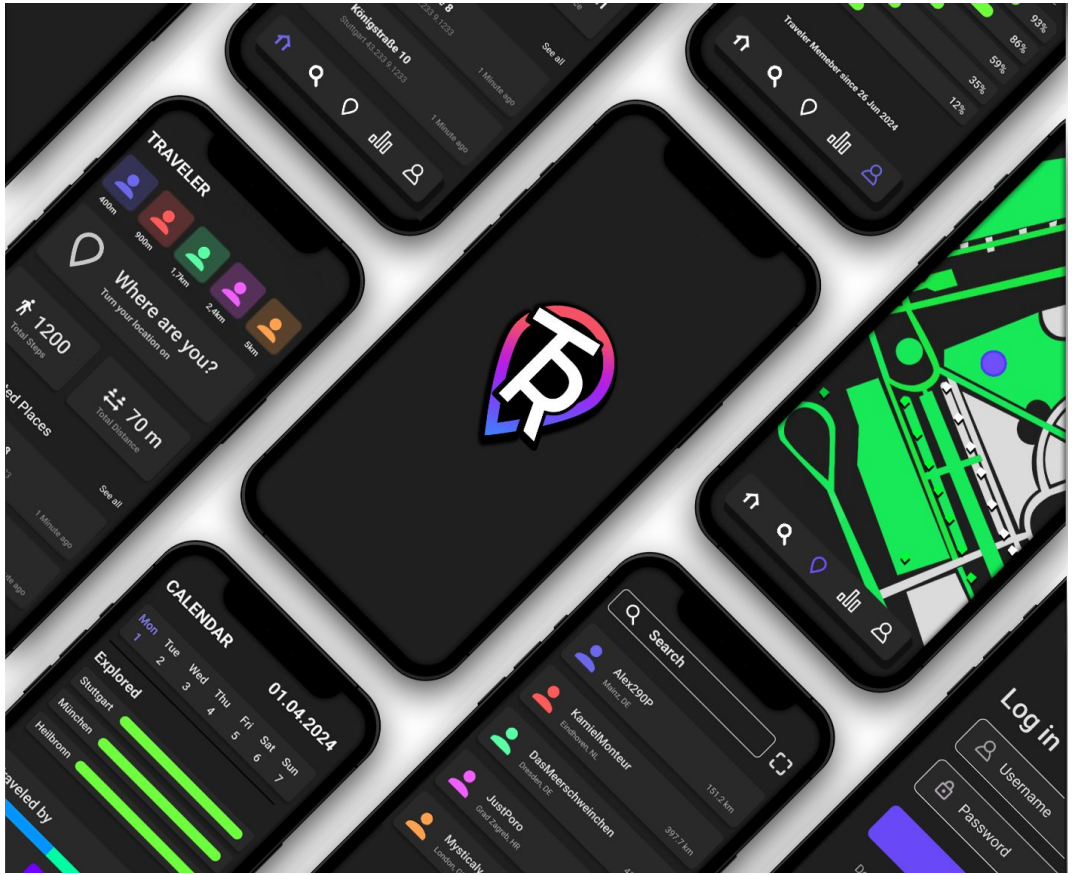
JUNI 2024

AN **ANDROID APP** DESIGNED TO GET **PEOPLE OUTSIDE** BY USING **REAL-TIME LOCATION** IN A **3D WORLD**.

IT ENCOURAGES **EXPLORATION** AND CONNECTION IN **REAL LIFE** BY KEEPING A RECORD OF **AREAS EXPLORED**, HELPING USERS **DISCOVER NEW PLACES**.

BACHELOR'S THESIS

GRADE 1.0



MOTIVATION AND CONCEPT IDEA

THIS IS WHERE **GAMING** MEETS **SOCIAL MEDIA**

THE APP IS DESIGNED AS A RESPONSE TO **STRENGTHEN THE CONNECTION** BETWEEN **VIRTUAL AND PHYSICAL** WORLDS.

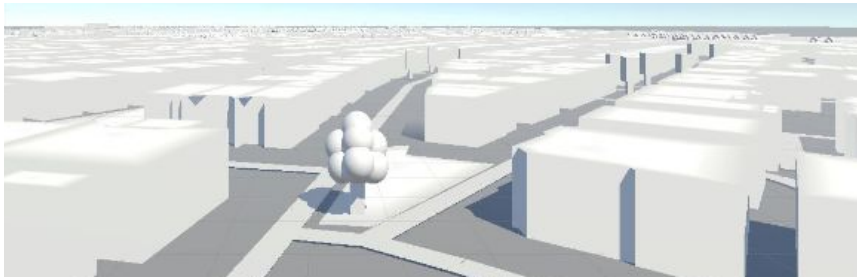
THE AIM IS TO **INSPIRE USERS TO BECOME ACTIVE** AND **EXPERIENCE** PERSONAL **ADVENTURES**, RATHER THAN LOSING THEMSELVES IN VIRTUAL REALITIES.

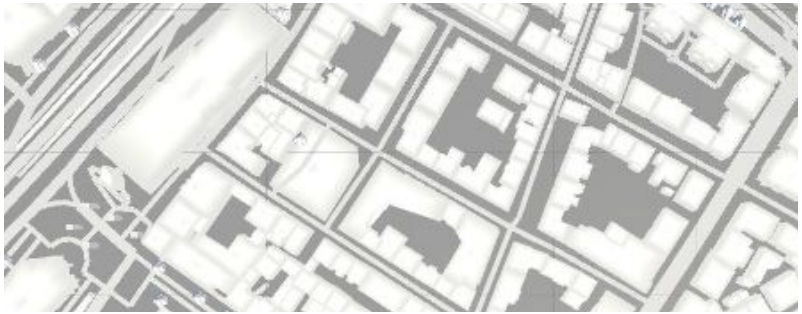
BUILT WITH **FLUTTER** FOR A SLEEK AND **USER-FRIENDLY UI**, AND POWERED BY **UNITY** FOR **HIGH-PERFORMANCE 3D RENDERING**, THE APP FEATURES AN IMMERSIVE STREET VIEW-BASED MAP THAT LETS **USERS** SEE **THEIR LOCATION** AND THAT OF **THEIR FRIENDS** IN **REAL TIME.** THIS SEAMLESS INTEGRATION CREATES AN **EASY-TO-USE EXPERIENCE** THAT BRINGS **EXPLORATION AND CONNECTION** TO LIFE.



UNITY X OPENSTREETMAP

THE **UNITY PART** OF THIS PROJECT, SENDS A **REQUEST** TO THE **OPENSTREETMAPS API** TO RETRIEVE **GEOGRAPHIC DATA** FOR THE USERS LOCATION IN ORDER **TO BUILD** THE RELEVANT **MAP OBJECTS**. SUCH AS **BUILDINGS**, **ROADS** AND **TREES**. THESE ELEMENTS ARE THEN PROCESSED AND CONVERTED INTO **3D OBJECTS**, MATCHING THEIR **REAL-WORLD POSITIONS** AND DIMENSIONS. THE FOLLOWING SCREENSHOTS SHOW A PART OF THE TRAIN STATION IN STUTTGART.

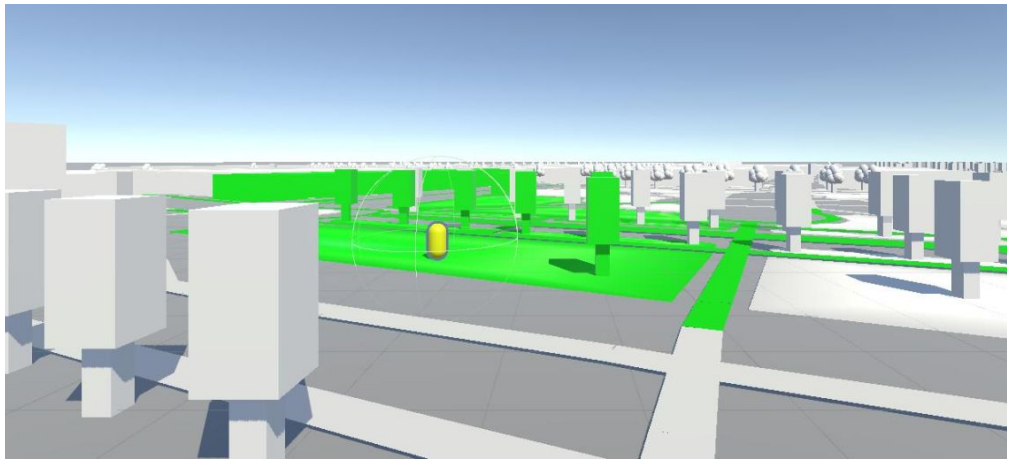


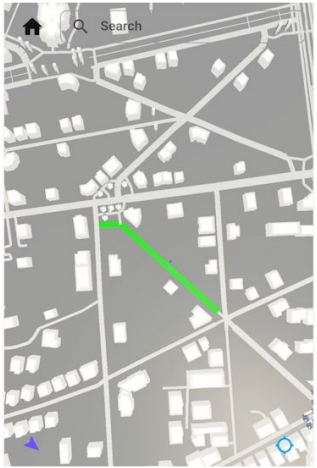


THIS SYSTEM **ALLOWS** THE CREATION OF A **DYNAMIC 3D WORLD** THAT REFLECTS THE REAL ENVIRONMENT. **USERS** CAN **NAVIGATE** THROUGH THIS WORLD **USING GPS** OR OTHER POSITIONING METHODS, ENABLING IMMERSIVE AND INTERACTIVE EXPERIENCES **BASED ON** THEIR ACTUAL **LOCATION**.

UNITY X OPENSTREETMAP

THE **USER** (REPRESENTED IN YELLOW) HAS A **CIRCULAR HITBOX**. WHEN THE **HITBOX COLLIDES** WITH A **BUILDING**, **STREET**, OR **TREE**, THE **OBJECT TURNS GREEN**, INDICATING IT HAS BEEN **'VISITED'**. THE SCREENSHOTS BELOW DEMONSTRATE HOW THIS FEATURE LOOKS—WITH THE UNITY RENDERING ON THE LEFT AND THE FINAL IMPLEMENTATION IN THE APP ON THE RIGHT.





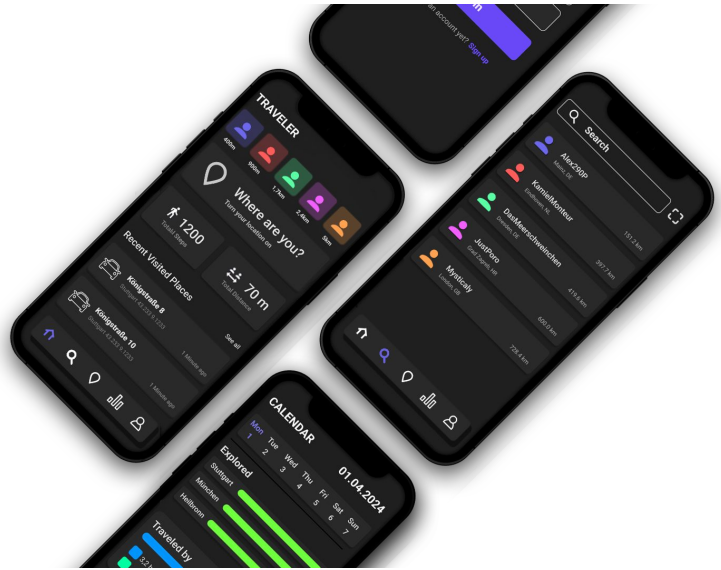


UI & UX IN FLUTTER

THE **FLUTTER** PART OF THIS PROJECT **SEAMLESSLY INTEGRATES** THE UNITY MAP COMPONENT, ENSURING A **SMOOTH** AND **INTUITIVE** USER EXPERIENCE.

FLUTTER BECAME THE BETTER CHOICE, SINCE IT IS AN OUTSTANDING FRAMEWORK FOR BUILDING ADAPTIVE, PERFORMANT AND **VISUALLY APPEALING** UI'S. ITS FLEXIBILITY ALLOWED ME TO CREATE A HIGHLY **RESPONSIVE DESIGN** THAT SCALES ACROSS DIFFERENT DEVICES WHILE MAINTAINING A **CONSISTENT LOOK** AND FEEL.

WITH FLUTTER'S POWERFUL WIDGET SYSTEM AND SMOOTH ANIMATIONS, I WAS ABLE TO DELIVER A POLISHED UI THAT **SEAMLESSLY CONNECTS WITH** THE **UNITY-POWERED** 3D WORLD, CREATING A COHESIVE AND IMMERSIVE EXPERIENCE.



WHAT THIS APP CAN DO

KEY FEATURES

REAL-WORLD 3D NAVIGATION: VIEW YOUR LOCATION IN A REAL-WORLD 3D ENVIRONMENT.

INTERACTIVE EXPLORATION: THE WORLD STARTS IN WHITE AND WHEN YOUR HITBOX TOUCHES OBJECTS, THEY TURN GREEN—MARKING THEM AS VISITED.

VISIT TRACKING & PROGRESS: YOUR VISITED LOCATIONS ARE SAVED IN THE BACKEND, ALLOWING YOU TO TRACK WHEN AND WHERE YOU'VE BEEN.

SOCIAL FEATURES: SEARCH FOR OTHER USERS, ADD THEM AS FRIENDS AND CONNECT.

CITY EXPLORATION STATS: SEE HOW MUCH OF A CITY YOU'VE EXPLORED, DISPLAYED AS A PERCENTAGE.

USER ACCOUNTS: SIGN UP, LOG IN, AND MANAGE YOUR PERSONAL ACCOUNT.

FOREST

JUNE 2023

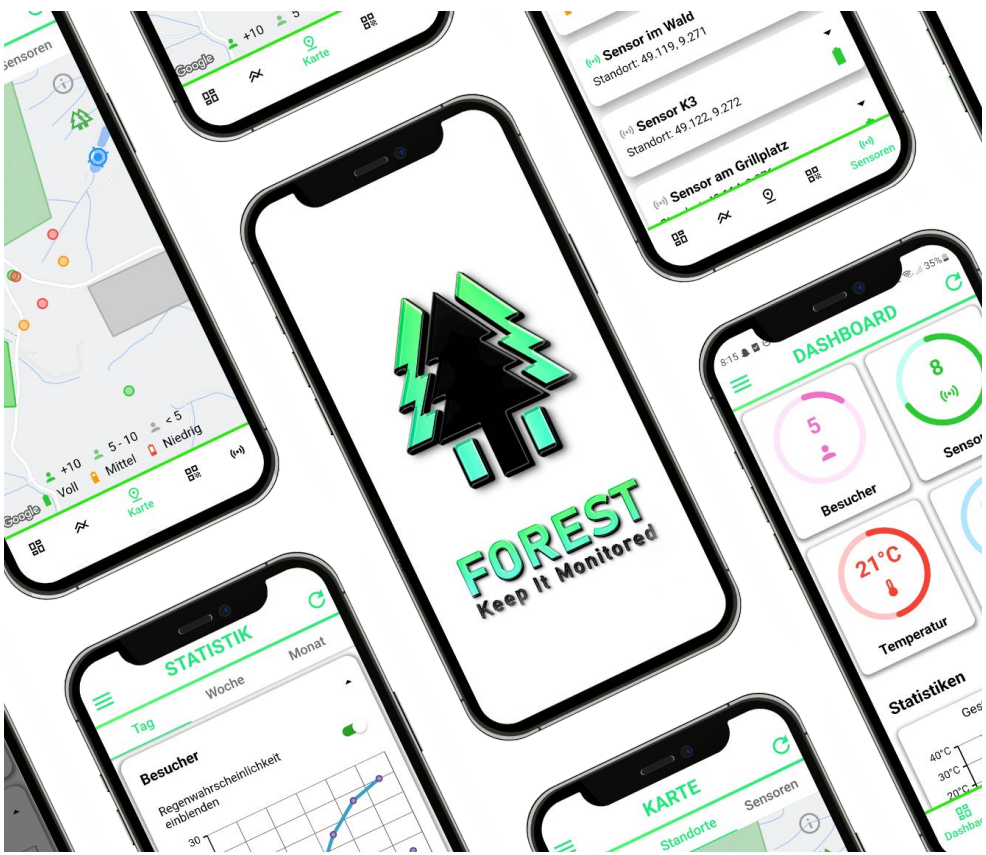
MONITOR YOUR VINEYARD

WITH THE IDEA TO MAKE THE **FOREST** IN HEILBRONN **SMARTER** WITH ADVANCED TECHNOLOGIES LIKE **THERMAL IMAGING CAMERAS** AND **VARIOUS SENSORS**.

THIS APP IS MADE **FOR EMPLOYEES** TO **MONITOR VINEYARDS**, **TRACK SENSOR LOCATIONS**, **CHECKING** THEIR **STATUS** AND **SEGMENT FOREST AREAS** TO MONITOR VISITOR MOVEMENTS.

MOBILE DEVICES

GRADE 1.3





GOOGLE MAPS

THE APP INCORPORATES **GOOGLE MAPS** TO DISPLAY **VARIOUS SECTIONS** OF A FOREST, ALLOWING USERS TO **VISUALLY IDENTIFY** THE NUMBER OF **PEOPLE IN** EACH **AREA** BASED ON COLOR CODING. BY **CLICKING** ON A **LOCATION OR AREA**, USERS CAN ACCESS **DETAILED DATA** RELATED TO THAT SPECIFIC SPOT. IT ALSO SHOWS THE **BATTERY STATUS** OF **SENSORS** WITHIN THE FOREST.

USERS CAN **TOGGLE** THE **VIEW** TO DISPLAY ONLY **SENSOR-EQUIPPED** AREAS OR ALL **AREAS** AS DESIRED. ADDITIONALLY, THE APP ENABLES **CUSTOMIZATION** OF **MULTIPLE FORESTS** AND PROVIDES THE ABILITY TO **QUICKLY NAVIGATE** TO **DIFFERENT LOCATIONS,** SUCH AS STUTTGART, ON THE MAP.

USER INTERFACE & USER EXPERIENCE

THE **USER INTERFACE** AND EXPERIENCE OF THIS APP ARE SPECIFICALLY **DESIGNED** FOR **GERMAN-SPEAKING** VINEYARD OWNERS, TAKING INTO ACCOUNT AN **OLDER TARGET AUDIENCE**.

TO ENHANCE **READABILITY** AND **ACCESSIBILITY**, THE DESIGN FEATURES **LARGE FONT** SIZES AND PRESENTS INFORMATION IN A COMPACT FORMAT, FOCUSING ON THE MOST ESSENTIAL DATA. THIS APPROACH ENSURES THAT THE APP IS USER-FRIENDLY AND EFFICIENT, PROVIDING A SEAMLESS EXPERIENCE THAT IS BOTH PRACTICAL AND **EASY TO NAVIGATE**.

BY **MINIMIZING COMPLEXITY,** THE APP AIMS TO EMPOWER USERS TO MAKE INFORMED **DECISIONS QUICKLY** AND **EASILY**, ENHANCING THEIR OPERATIONAL EFFICIENCY AND OVERALL SATISFACTION.



WHAT THIS APP CAN DO

KEY FEATURES

LIVE VISITOR COUNT: SEE THE CURRENT NUMBER OF VISITORS IN REAL TIME.

REAL-TIME ENVIRONMENT DATA: VIEW LIVE TEMPERATURE AND ACTIVE SENSOR READINGS.

DETAILED SENSOR INSIGHTS: FILTER SENSORS, ADD NEW ONES VIA GOOGLE MAPS AND ACCESS DETAILED DATA PER SENSOR.

CUSTOM FOREST MANAGEMENT: DIVIDE YOUR FOREST INTO SECTIONS, ADD NEW FORESTS AND SWITCH BETWEEN LOCATIONS.

INTERACTIVE DATA ANALYSIS: VISUALIZE YOUR DATA THROUGH DETAILED STATISTICS AND TRENDS.